Quiz 2 (pp. 57)

September 27, 2023

1

$4x^2 = 64$
$\rightarrow x^2 = 16$
$\rightarrow x = \pm 4$

2

$3(p-1)^2 = 15$
$\rightarrow (p-1)^2 = 5$
$\rightarrow p - 1 = \pm\sqrt{5}$
$\rightarrow p = 1 \pm \sqrt{5}$

3

$16(m+5)^2 = 8$
$\rightarrow (m+5)^2 = \frac{1}{2}$
$\rightarrow m + 5 = \pm\frac{1}{2}$
$\rightarrow m = -5 \pm \frac{\sqrt{2}}{2}$

4

$-2z^2 = 424$
$\rightarrow z^2 = -212$
$\rightarrow z = \pm\sqrt{-212}$
$\rightarrow z = \pm 2\sqrt{53}i$

5

$s^2 + 12 = 9$
$\rightarrow s^2 = -3$
$\rightarrow s = \pm\sqrt{-3}$
$\rightarrow s = \pm\sqrt{3}i$

6

$7x^2 - 4 = -6$
$\rightarrow 7x^2 = -2$
$\rightarrow x^2 = -\frac{2}{7}$
$\rightarrow x = \pm\sqrt{-\frac{2}{7}} = \pm\frac{\sqrt{14}}{7}i$

7

$(5 - 3i) + (-2 + 5i)$
$\rightarrow 3 + 2i$

8

$(-2 + 9i) - (7 + 8i)$
$\rightarrow -9 + i$

9

$3i(7-9i)$
$\rightarrow -27i^2+21i$
$\rightarrow 27+21i$

10

$(8-3i)(-6-10i)$
$\rightarrow -48-10i+16i+30i^2$
$\rightarrow -48-30+6i$
$\rightarrow -78+6i$

11

$\frac{4i}{-6-11i}$
$\rightarrow \frac{4i(-6+11i)}{(-6-11i)(-6+11i)}$
$\rightarrow \frac{-44-24i}{157}$
$\rightarrow -\frac{44}{157}-\frac{24}{157}i$

12

$\frac{3-2i}{-8+5i}$
$\rightarrow \frac{(3-2i)(-8-5i)}{(-8+5i)(-8-5i)}$
$\rightarrow \frac{-34+i}{89}$
$\rightarrow -\frac{34}{89}+\frac{1}{89}i$

13

$y=x^2-4x+9$
$\rightarrow y-9=x^2-4x$
$\rightarrow y-5=x^2-4x+4$
$\rightarrow y-5=(x-2)^2$
$\rightarrow y=(x-2)^2+5$, and thus, the vertex is $(2,5)$.

14

$y = x^2 + 14x + 45$
$\rightarrow y - 45 = x^2 + 14x$
$\rightarrow y + 4 = x^2 + 14x + 49$
$\rightarrow y + 4 = (x + 7)^2$
$\rightarrow y = (x + 7)^2 - 4$
$\rightarrow$, and thus, the vertex is $(-7, -4)$.

15

$f(x) = x^2 - 10x + 17$
$\rightarrow f(x) - 17 = x^2 - 10x$
$\rightarrow f(x) + 8 = x^2 - 10x + 25$
$\rightarrow f(x) + 8 = (x - 5)^2$
$\rightarrow f(x) = (x - 5)^2 - 8$, and thus, the vertex is $(5, -8)$.

16

$g(x) = x^2 - 2x - 7$
$\rightarrow g(x) + 7 = x^2 - 2x$
$\rightarrow g(x) + 8 = x^2 - 2x + 1$
$\rightarrow g(x) + 8 = (x - 1)^2$
$\rightarrow g(x) = (x - 1)^2 - 8$, and thus, the vertex is $(1, -8)$.

17

$y = x^2 + x + 1$
$\rightarrow y - 1 = x^2 + x$
$\rightarrow y - 0.75 = x^2 + x + \frac{1}{4}$
$\rightarrow y - 0.75 = (x + 0.5)^2$
$\rightarrow y = (x + 0.5)^2 + 0.75$, and thus, the vertex is $(-0.5, 0.75)$.

18

$y = x^2 + 9x + 19$
$\rightarrow y - 19 = x^2 + 9x$
$\rightarrow y + 1.25 = x^2 + 9x + 20.25$
$\rightarrow y + 1.25 = (x + 4.5)^2$
$\rightarrow y = (x + 4.5)^2 - 1.25$, and thus, the vertex is $(-4.5, -1.25)$.

19

When an object is droped, its height h (in feet) above the ground after t seconds can be modeled by the function:
$h = -16t^2 + h_0$
where h_0 is the object's initial height (in feet) (see pp. 34). Thus, the equation modeling the given situation is $h = -16t^2 + 45$. To solve this question, we need to find the roots of the above equation:
$0 = -16t^2 + 45$
$\rightarrow -16t^2 = -45$
$\rightarrow t^2 = \frac{45}{16}$
$\rightarrow t = \pm\frac{\sqrt{45}}{\sqrt{16}} = \pm\frac{3\sqrt{5}}{4}$. Because only positive times are accepted, $t = \frac{3\sqrt{5}}{4}$, which is how long the ball is in the air for.